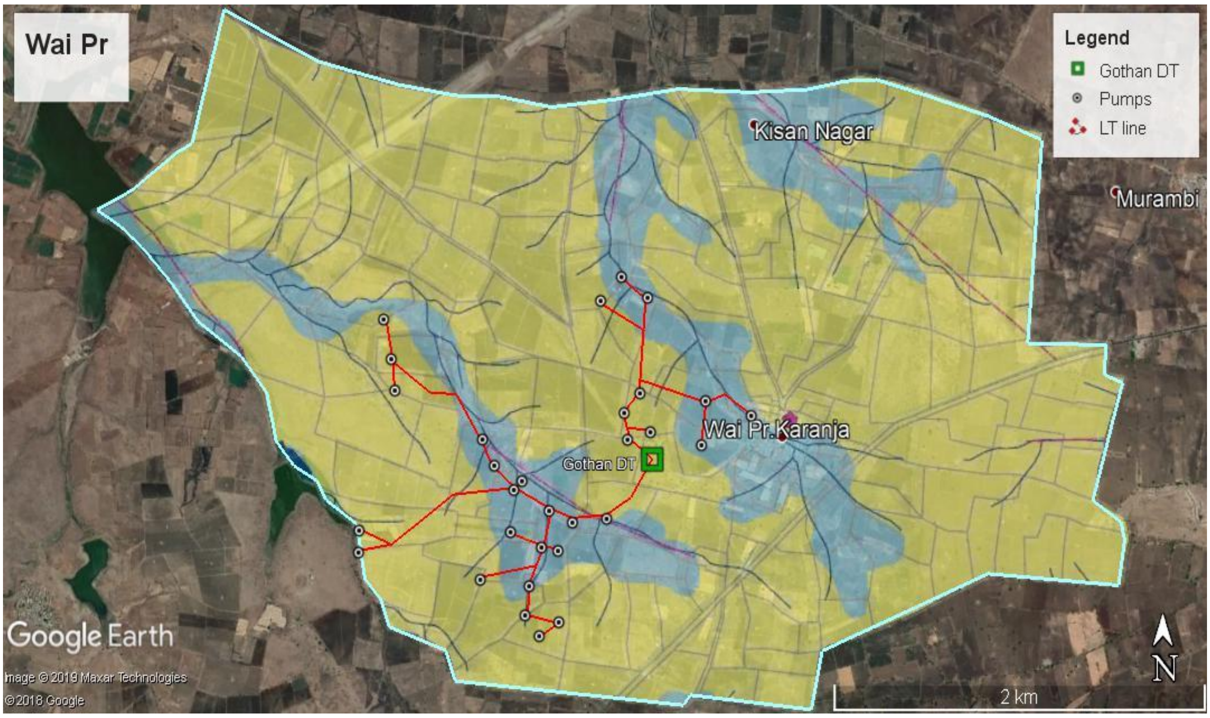
Analysis of Energy Infrastructure for Irrigation: LT network restructuring and other approaches for improved supply



This study was submitted as a part of the MoU III between IIT Bombay and the Project on Climate Resilient Agriculture (PoCRA), Dept. of Agriculture, GoM

The GIS documentation and restructuring methodology was done by Prathamesh Antarkar as a part of his M.Tech. project in the Centre for Technology Alternatives for Rural Areas, IIT Bombay

September, 2020

Table of Contents

List of Figures

List of Tables

1. Electricity Infrastructure for Irrigation in Umbarda Village

1.1. Introduction

An important objective of the Project on Climate Resilient Agriculture (PoCRA) is to increase the on-farm availability of surface water and groundwater for protective irrigation to deal with the effects of erratic rainfall. The increased reliance on energized irrigation coupled with shortfall in energy infrastructure capacity intensifies the uncertainty. Field visits and other observations from previous research indicate electricity quality and access being a hindrance to productivity and sometimes a cause for excess water usage. Umbarda village in Karanja, Washim has been selected for these investigations.

In this work we have come up with a restructuring approach to improve voltage profiles and reduce potential overloading on Distribution Transformers. Since the LT network has developed over the years with the increasing demand for connections, the final layout is unlikely to be optimal. Hence there is an opportunity to restructure the network by moving the LT lines so as to make an optimal design.

Additionally, many of the electricity issues faced by farmers such as low voltages, pump tripping, and DT overloading, are found to arise from a latent demand which includes pending connections and unsanctioned loads. The restructuring exercise can include adding new connections. Hence the result is new connections, and an improved profile, at a cost which is substantially lower than the average cost of connection, and lower maintenance costs for MSEDCL (for e.g. due to DT failure).

The methodology and heuristics developed in this report can be implemented in other villages too.

The contents of the report are as follows:

- Documentation of the grid infrastructure present in Umbarda in GIS
- Restructuring of the LT network to add new connections at minimal cost and improve the voltage profile
- Identification of new connections that can be added by an extension of an LT line by up to 3 poles and without addition of DTs
- A few other proposed optimizations that can reduce infrastructure costs to MSEDCL and satisfy latent demand

1.2. Village background

Umbarda village has 189 sanctioned consumers and 14 Paid Pending consumers. However, from our field observations, we find that the latent demand for connections is usually much higher.

Table 1: Umbarda Village Overview

Population:	**4933**	**Number of farmers:**	**893**
Number of electricity connections:	**189**	**Number of DTs:**	**15**
Number of paid pending:	**19**	***No. of unconnected wells:**	**71**
Cultivable area of village:	**1300 ha**		

* Mapped by using the satellite images excluding the wells outside village boundary.

To make some projections of what solutions are possible, and develop a methodology which could be used in all villages, we have assumed that the latent demand is due to the unconnected wells observed in satellite images and the paid pending connections. In discussions with wiremen and some farmers, we found that 70-80% of these wells may be active. All costing is based on MSEDCL Proposed Costs data 2019-20. ((Costs in Appendix II)).

The main outcomes of this analysis are:

- The average electricity infrastructure cost per connection for sanctioned consumers is Rs. 1,10,962/- in Umbarda. This is the average cost per consumer considering the existing infrastructure of substation, HT lines, LT lines and DTs in the village. The high tension infrastructure is averaged over all the connections on the feeder/substation for the feeder and substation costs respectively. This average cost is almost the same as MSEDCL's average over 6 years in the state: Rs. 1 lakh per connection.
- The network has developed over time to address each new connection, resulting in a haphazard structure in places. We analysed the value of making changes to the older network while also connecting an aggregate set of the new loads. We found that 70 new connections could be made on the stretch of the two feeders in Umbarda village at an average cost of Rs. 26,275/- per connection plus the cost of moving LT lines of about 11 km length. This is substantially lower than Rs. 62,664 /- the cost that would be incurred had the current MSEDCL procedure been followed. Additionally, the voltage profile on the system has improved.
- Within the current MSEDCL process, we try to understand how many farmers could potentially get the requisite connections at a significantly lower than average cost.

Out of the unsatisfied demand, 31 connections can be made without the addition of any Distribution Transformer with an input cost ranging from Rs. 1600 to Rs. 40,819, with 10 requiring only a service line, and 11 requiring 1 to 2-pole extensions, and hence an expenditure of less than ~Rs. 25,000/-.

- Current substation capacity allows 704 connections to be added. Hence up to 37% such new connections could potentially be added in the 9 villages supplied by Umbarda substation.
- While all of this analysis is based on proxy loads, it does lay the ground work for a new approach to electricity access and better quality of power. This is a new approach which could lead to benefits for all stakeholders. A set of heuristics which can be repeated in any village, have also been developed.

1.3. Methodology for the study

The approach in this study was to select one village typical of Vidarbha and Marathwada with a cropping pattern of predominantly Soybean, Tur, and Cotton, where problems related to the quality of electricity exist, and could be analysed further. The village was selected based on secondary data and preliminary knowledge about the existing issues regarding problems of electricity in the villages of Vidarbha and Marathwada. A detailed analysis of a distribution network of a village is done to address the technical issues being faced by farmers as a case study that can be generalized further.

1.3.1 Selection of location for field work

The selection of the village was based on the following criteria.

- The village should be a PoCRA village
- The village should have problems related to overloading, transformer and pump failure (which is reflective of voltage issues).
- The village should be closer to a substation for effective implementation of load management strategies

The purpose of the project is to understand and analyse the problems related to overloading. The village having problems related to overloading, transformer failure, and pump burnout would help to explore the possibility of improving the distribution network and analyse possible effects of load management strategies with given constraints.

1.3.2. Data collection

The primary data collection involved a focused group discussion conducted with the farmers in the identified village. The focused group discussions with the farmers gave an insight into the problems associated with the quality of electricity supply. The discussion helped to identify the transformers in the village that had failed frequently in the last few years. Sample measurements of voltage and power drawl were carried out at different transformers to know the actual load on the network. The single line diagram of all DTs was obtained from the local wireman for further analysis. A detailed interview of the Assistant Engineer, Subdivision Karanja Lad, was conducted to understand the energy consumption pattern over the year, various practices implemented by MSEDCL to handle overloading, particularly in the peak season, and design process followed for new connections.

The secondary data related to electricity infrastructure, transformer failure data, consumer data, and energy consumption data at the feeder level was collected from Umbarda substation and sub-division office in Karanja Lad. Rainfall data from maharain, and annual water budget data, and cropping data was collected from PoCRA, to understand the scenario of water sources and water usage in irrigation.

The analysis of secondary data about transformer failure helped to identify a village that satisfied all the mentioned criteria. Karanja block of Washim district was selected for convenience as well as a typical agro-climatic area. The detailed analysis of the data collected, existing scenario and recommendations for improvement are given in subsequent sections.

2. Description of Electricity Infrastructure in Umbarda

Umbarda village is supplied by two feeders, Umbarda and Somthane, emanating from Umbarda substation. A quick overview of the electricity infrastructure and loading information is given in Table 2, the electricity distribution infrastructure of all the 15 DTs in Umbarda Village is mapped in Figure 1.

Table 2: Distribution infrastructure overview of subdivision

	No. of Sanctioned Consumers	Connected Load (kW)	No. of farmers in Paid Pending list	% Paid Pending
Umbarda Village	189	666	17	9
Umbarda Ag Feeder	1386	4073	103	8
Somthane Ag feeder	505	1475	24	5
Umbarda Substation	1925	5645	128	7
Karanja Subdivision	11324	31414	1051	9

In Umbarda Village there are 8% paid pending connections, from our understanding of the applications process, there are likely to be more applicants that have not reached the paid pending stage. The actual number of unsanctioned loads operational during the peak season are close to 80% of the all unconnected and operational wells in and around Umbarda.

Per farmer basic electricity infrastructure cost for the existing connections on each Distribution Transformer in Umbarda village is calculated and shown in Table 3.

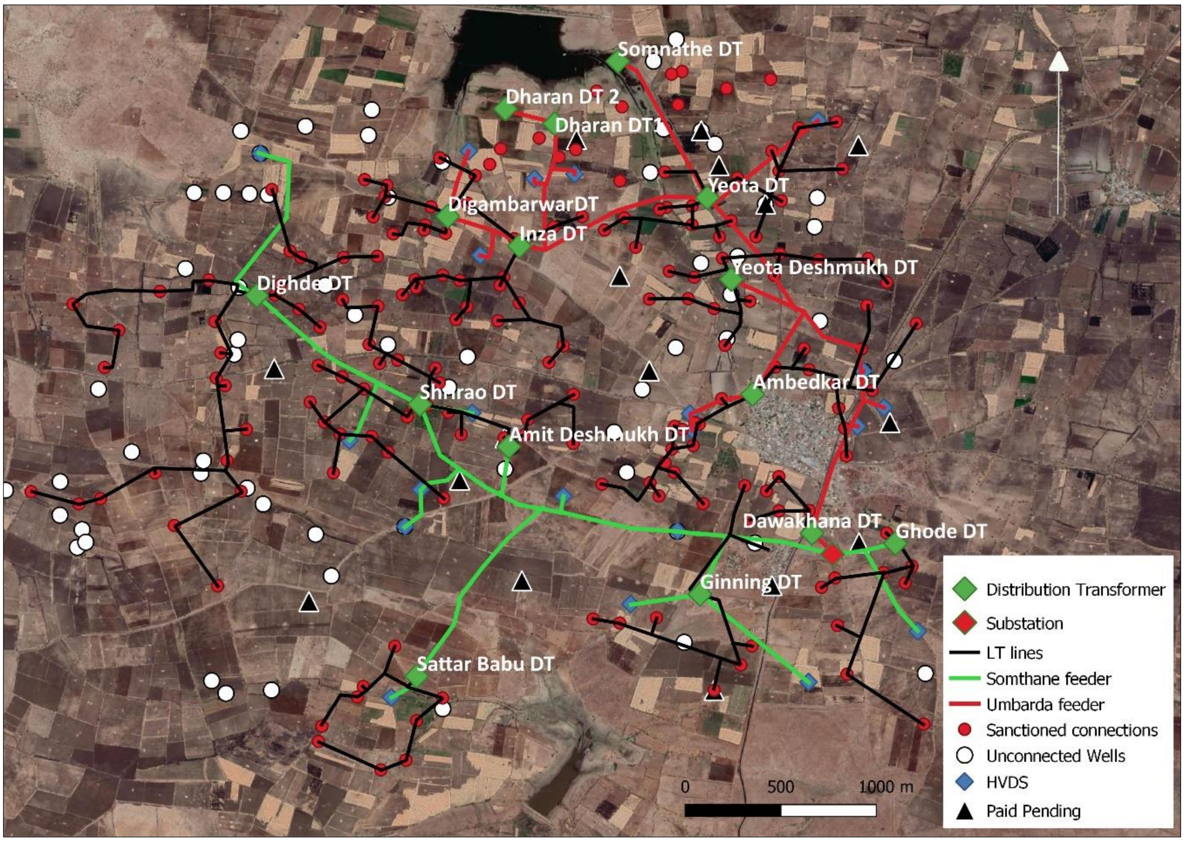


Figure 1: Umbarda village existing electricity infrastructure

Table 3: Cost per farmer: Umbarda Sanctioned Consumers

Sr. No.	DT name	LT line length (km)	No. of Sanctioned Consumers	Total cost per connection (HT+ LT + substation) (Rs.)
1	Ambedkar	4.3	25	81,245
2	Yevta Deshmukh	2.5	13	1,10,425
3	Yevta Road	3.2	19	88,005
4	Somnathe	0	6	1,19,177
5	Inza	2.8	18	86,420
6	DigambarVar	1.6	13	91,586
7	Dharan 1	0	5	1,41,476
8	Dharan 2	0	5	1,41,476
9	Dawakhana	0.8	5	1,85,016
10	Ghode	2.0	9	1,40,971
11	Ginning (63kVA)	2.6	7	2,13,663
12	Sattar Babu	2.0	9	1,55,084
13	Amit Deshmukh	2.7	16	1,01,650
14	Shrirao	2.1	12	1,20,501
15	Dighde	6.7	27	1,03,681

**All are 100kVA DTs except Ginning DT; The HP rating for consumers is not available yet.*

- Average cost per connection is Rs. 1,10,962 /-, ranging between Rs. 81,245 /- and Rs. 1,87,915 /- separately for each DT.
- Rs. 22,526 /- (21%) is the distribution transformer cost per farmer Rs 51,829 /- (48%) is the per farmer LT line cost, Rs. 26,877 /- (25%) is the per farmer HT line cost and a 7% of cost is for the substation infrastructure. The major part of the cost is for LT lines.
- The average LT line length is 2.7 km per DT.
- The DTs near the Dharan have been excluded in the average LT line length calculation, because there is no LT line, but farmers are made to draw their own cables from the DT to their pumps. (See Appendix IV)
- The per farmer infrastructure cost in Table 3 is calculated considering only the sanctioned connections. Perhaps there would be additional connections as well, which might include the farmers in the paid pending list as well as other unconnected wells and bore wells in the region.

2.1. Additional connections possible on current DTs

In this section we attempt to identify the low hanging fruit for connection allocations: we identify water sources than can be connected by extension of LT lines without overloading any of the existing DT, and without adding new DTs.

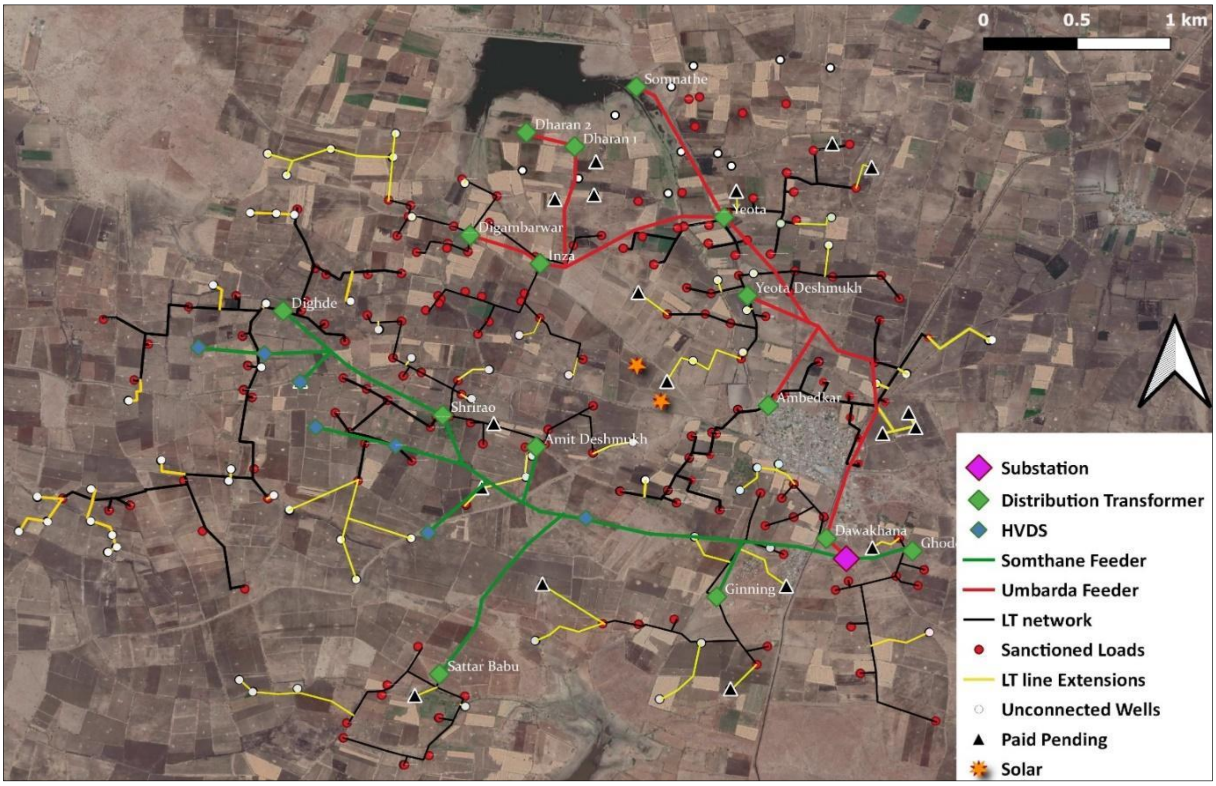


Figure 2: Distribution network after adding unconnected loads

The yellow lines in Figure 2 indicate these extensions. All extensions follow farm boundaries. Table 4 indicates the cost of extension for these connections. This is in comparison to an average cost for the existing connections of Rs 1,10,962/- in Umbarda, and almost the same average cost across the state of Rs. 1 lakh.

Table 4: Cost of connection for loads on existing DT

No. of poles extension*	No. of connections	Cost (Rs.)
Service line only	10	1,600
1	6	13,606
2	5	27,212
3	6	40,819
4 to 9	16	76,927 (Avg.)

**Each pole extension is for 50 m distance*

The additional per farmer cost to connect the closest 27 water sources, or 27% of the pending proxy loads, is Rs. 17,000/- per connection, and the closest 43, or 43% of the pending proxy loads, is Rs 40,000/- per connection, on an average.

In this section we have assumed that a maximum 24 of 5 HP loads can be connected on a 100 kVA transformer since that is the highest number of connected loads in Umbarda are 25 and 27, on Ambedkar and Dighde DTs respectively. The maximum connected load in fact depends on the load diversity factor which is discussed in detail in the next section.

2.2. Load Diversity Factor

The number of pumps that can be accommodated on a transformer depends on the load diversity factor, which is defined as the ratio of the sum of maximum demands of individual loads connected to the system to the simultaneous maximum demand:

$$f_{Diversity} = \frac{\sum_{i=1}^{n} Individual\ peak\ load}{Maximum\ Demand} = (In\ the\ case\ of\ irrigation\ pumps) = \frac{\sum_{i=1}^{n} Connected\ Load}{Maximum\ Demand}$$

A higher Load Diversity Factor results when there is a greater variation in the irrigation patterns of the farmers connected to a DT. The minimum Load Diversity Factor is 1, which means that there is an occurrence of all farmers using their pumps at the same time.

There is uncertainty in the load diversity factor, and it is hard to gauge what the maximum number of consumers that can be accommodated on a DT due to two reasons. One is that while we know the sanctioned loads on a DT, it is unclear how many unsanctioned loads are operating from that same DT. In addition since there are no meters on DT, there is no record of the loading on any DT. We have made some estimates based on (a) the average irrigation requirements of the farmers, (b) the maximum connected loads on the DTs in the village, and (c) the ratio of maximum power drawl to connected load on the feeder - maximum feeder demand is available.

The load diversity factor is an important number to help determine the maximum number of loads that can operate on a DT without allowing failure. MSEDCL has a few different standards as regards this number. From an office in Yavatmal we find that a 100 kVA DT should have a maximum of 107 HP connected load, which is an assumption of a load diversity of 1.25, planning for a maximum loading of 80 kVA on a 100 kVA DT. Umbarda feeder has an average 74 HP per DT. A 100 kVA DT in Wai Pr. (Karanja, Washim) has a connected load of 101 HP. In Walwa, Sangli we find an average connected load of 240 HP per 100 kVA DT on Bahe feeder, which is a load diversity of greater than 2.4, and in Niphad, Nasik, we find 100 HP on 100 kVA on a DT on Behed feeder. Regions where irrigation is given regularly the load diversity will be high as seen in the Sangli case. We still need to investigate how MSEDCL determines, and/or codifies this number. This is a critical parameter in design, planning, and cost optimization.

Considering the case of Ambedkar DT, 25 sanctioned consumers cover an area of 90.12 acres. Field measurements of 7 farmers with 3 and 5 HP pumps indicated flow rates of 22 to 45 m^3/hr with an average of 25 m^3/hr. Considering an irrigation depth of 5 cm, it takes 25 pumps 3.6 days to irrigate 90 acres. If we assume that all farmers want to irrigate their land within 4 days in case of a dry spell, then a 100 kVA DT could support at most 25 pumps of 4 HP load. Some load management may be required, since 25 pumps with an average load of 4 HP could lead to a somewhat greater power drawl than 100 kVA, but it is also possible that farmers are not irrigating/cultivating all of their land. Currently, DT power drawl is not being monitored or measured by MSEDCL, however the wireman indicated that 5-10 unsanctioned loads do access power from Ambedkar and Dighde DTs. Assuming an additional 5 loads at an average of 4 HP, we get a connected load of 120 HP. A loading of 80 kVA was observed on Inzha DT on 6 February 2020. Based on the above estimates and measurements, the Load Diversity is expected to be 1.2 to 1.5.

3. Restructuring the LT Network

In section 2.1 we analysed how many additional connections could be given by the extensions of 1 to 3 LT poles. In this section we explore a new approach to restructure the network and connect all pending potential consumers at minimal cost while ensuring good quality power supply to all. Water sources have been developing over a period of time, and the LT (and possibly HT) network has developed over time to address each new connection. This has resulted in the ad hoc and sometimes haphazard development of the network. In this section we analyse the value of making changes to the older network while also connecting the pending loads.

We interviewed a few Assistant Engineers at sub-division offices, to understand the process that MSEDCL follows in giving new connections. A field survey is conducted and the nearest LT pole is identified – the connection is made by extending the LT line from this point. The wireman decides if the concerned distribution transformer can handle the new load, and if so the connection is made to that DT. And if the DT is already loaded to its limit then it is connected to the next nearest DT. This process may lead to long LT lines. Interviews with other MSEDCL personnel also indicate that sometimes farmers are told to collect a few more requirements in the same area, and if they can, a new DT is added to give out all the connections.

While the connections are implemented in batches by contractors, it is not clear what design protocol is used by the contractor.

3.1. Restructured network

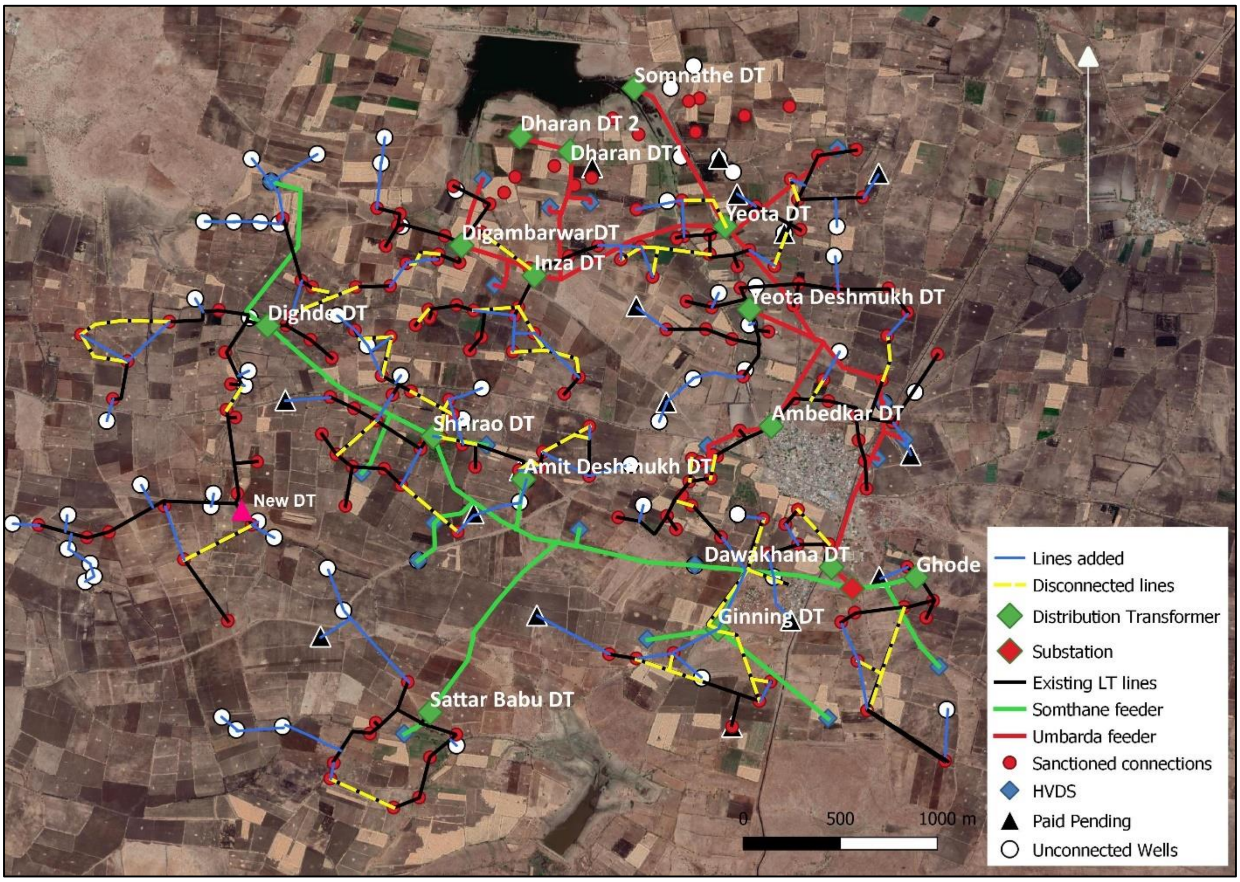


Figure 3: Proposed energy infrastructure. The LT lines to be disconnected are in yellow and the LT lines to be added are in blue. The one new DT is indicated as a pink triangle.

Figure 3 shows the restructured network that includes a connection to all potential loads. The light blue lines show new LT lines, the yellow lines are disconnected LT lines. The pink triangle in the South West corner of the village is one new DT.

3.2. Process followed while restructuring the LT network

i. All unconnected wells as seen from the satellite image are categorized into 3 categories: near Gaothan, seasonally used, unused or abandoned.

ii. The wells outside the village boundary and lying in the vicinity of the LT network of adjacent village are removed initially except for Dighde DT (as the LT network of Dighde DT is also present in Jamgaon village which is situated to the west of Umbarda). 21 such wells were removed.

iii. Following this, wells near Gaothan that are not utilized for agriculture purpose and unused or abandoned well were removed. 7 such wells from both categories were removed.

iv. In the next step, the wells outside 300 m distance from an already existing pole on LT network or a DT were removed. The 300 m distance has been set considering the maximum practical distance in order to get the desired voltage as found out from the in depth interviews with wireman and Assistant Engineer, Karanja Lad. 8 such wells were removed.

v. The remaining 56 wells were connected to the LT network as per the set of heuristics, developed to achieve the restructuring. The set of heuristics is listed in the appendix.

vi. Additionally, if a well is outside the 300 m radius circle from a pole on the LT network, but within the 300 m radius from an existing Paid Pending connection, then that well has been taken into account while restructuring the network.

vii. Apart from that, some of the older loads have been moved from two transformers which were already loaded to capacity – Ambedkar and Dighde DTs, to neighbouring DTs, so that some new loads could be connected to these two DTs.

viii. New LT lines were added, and some older lines were disconnected. The new structure ensured shorter LT lines for all loads, and none of the DTs getting overloaded.

ix. The restructuring resulted in proper voltages to all loads even when DTs were loaded to the maximum. This was tested through power flow simulations using MATLAB.

x. With the current set of connections, Ambedkar and Shirao DT have low voltage problems as informed by farmers during Focused Group Discussions. The power flow simulation results of Ambedkar DT network, for 100kVA and 80 kVA loading on the DT showed 186V & 194V at the farthest end of LT network respectively.

The exercise resulted in total 70 new connections over 189 existing connections, at an average cost of Rs.57,780/- per connection. This is only a little lower – Rs. 62,664 - than the cost that would be incurred had the current MSEDCL procedure been followed. However, we must note that no salvage value has been considered for the LT lines. The restructuring results in about 11 km of defunct and 16 km new LT lines. If the old LT lines are moved and re-used, the cost per connection is Rs. 26,275/- plus the cost of moving 11 km of LT lines. Additionally, the voltage profile on the system has improved.

Considering a more realistic scenario, from in depth interviews with farmers and lineman it was found that about 80% of the all unconnected and functioning wells in and around Umbarda is are in effective use during the peak season. Taking this into account, 59 new connections can be added at an average cost of Rs. 27, 484/- per connection plus the cost of moving 11 km of LT lines.

Since connections are implemented in aggregates in a village, an automated tool to do such planning could lead to a great deal of improvement and / or reduction in cost for MSEDCL. In future phases, a more realistic analysis of potential connections, reasons for unsanctioned consumers, and latent demand will be included.

3. Other recommendations

3.1. Optimizations to reduce infrastructure cost

As mentioned in the Introduction, agricultural feeders are a peculiar power supply system, with each feeder working only for 8 hours a day while being designed to meet peak loads. This is unlike other distribution systems which are also designed to meet peak loads, but can never be turned off since the consumers are always given the flexibility to draw power at any time. This means that the Ag feeders infrastructure can be utilized a lot more.

3.1.1. Switching on the LT Network

Maximum utilization of the existing grid infrastructure can be achieved by distributing loads over the day through the use of automated switches at the LT level. By operating the network for 16-24 hours instead of 8, the current infrastructure will be able to handle thrice the current loads. A complete solution will have to take into account technical aspects, usage constraints, farmers' habits regarding hooking, and variation in power drawl in the state.

3.1.2. Trade-off daytime supply for infra cost

Currently Ag feeders in Umbarda are operated in 2 time slots with each feeder getting 3 days of daytime supply and 4 days of night-time supply: 10 am - 6 pm, 10 pm - 6 am, avoiding the evening and morning peaks between 6 -10 am, and 6 – 10 pm. The Umbarda substation has a 5 MVA transformer which is sufficient to operate the 2 Ag feeders in this schedule, with the Gaothan feeder operating for 24 hours.

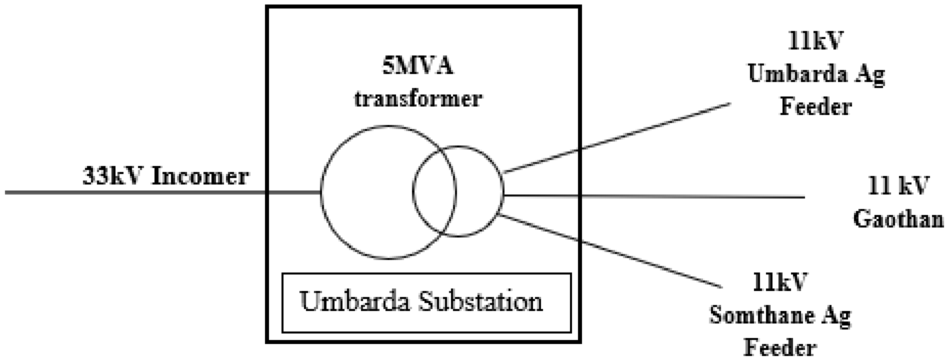


Figure 4: Schematic of feeders emanating from Umbarda Substation

But in the peak months of December and January, the load increases such that the Umbarda feeders current carrying capacity of 200 A is breached. So every year, the load on Umbarda and Somthane is split into three feeders which are called Manbha, Yevta and Somthane, with the split requiring a change only at the substation. Figure 5 shows the energy consumption over the year on Somthane and Umbarda. The split is made from Dec 4^{th} 2019 to Feb 4^{th} 2020. And up to Jan 2^{nd} in the previous year.

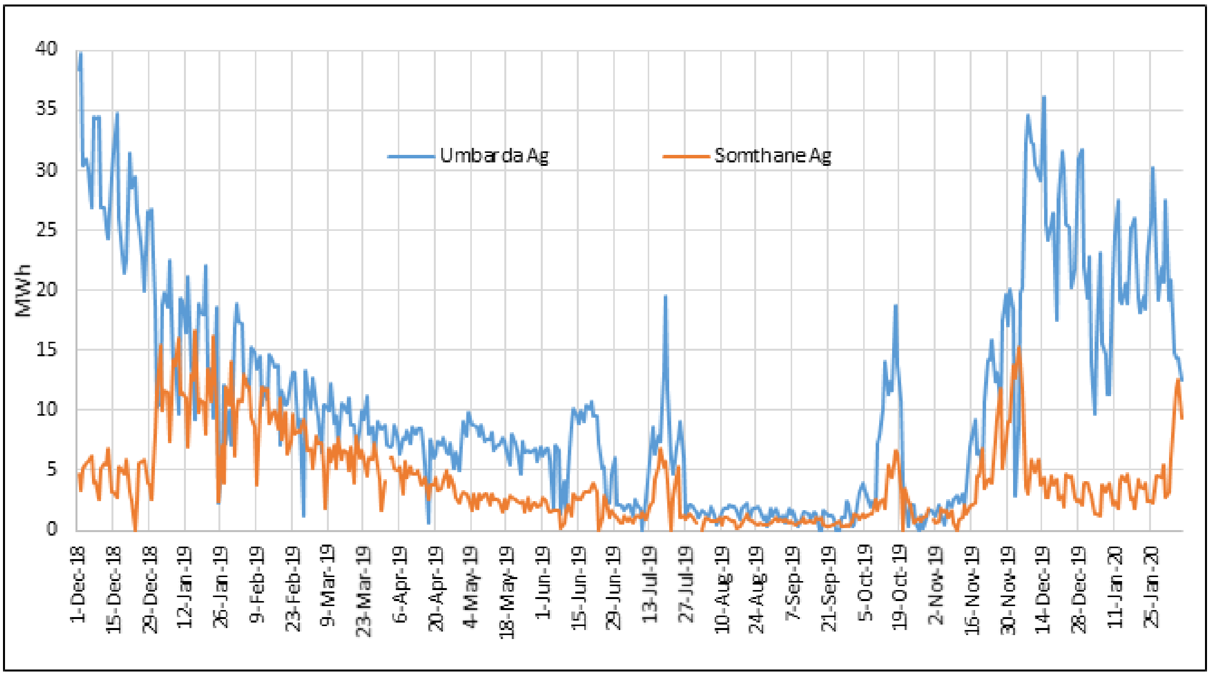


Figure 5: Daily energy Consumption (01 Dec 18 to 06 Feb 2020)

**The Umbarda feeder daily energy consumption is inclusive of Manbha and Yevta from Dec 4 to Feb 4*

But due to the heavy loading, all four feeders cannot be operated within the 16 hour period on a 5 MVA transformer. See the loading pattern in Figure 6.

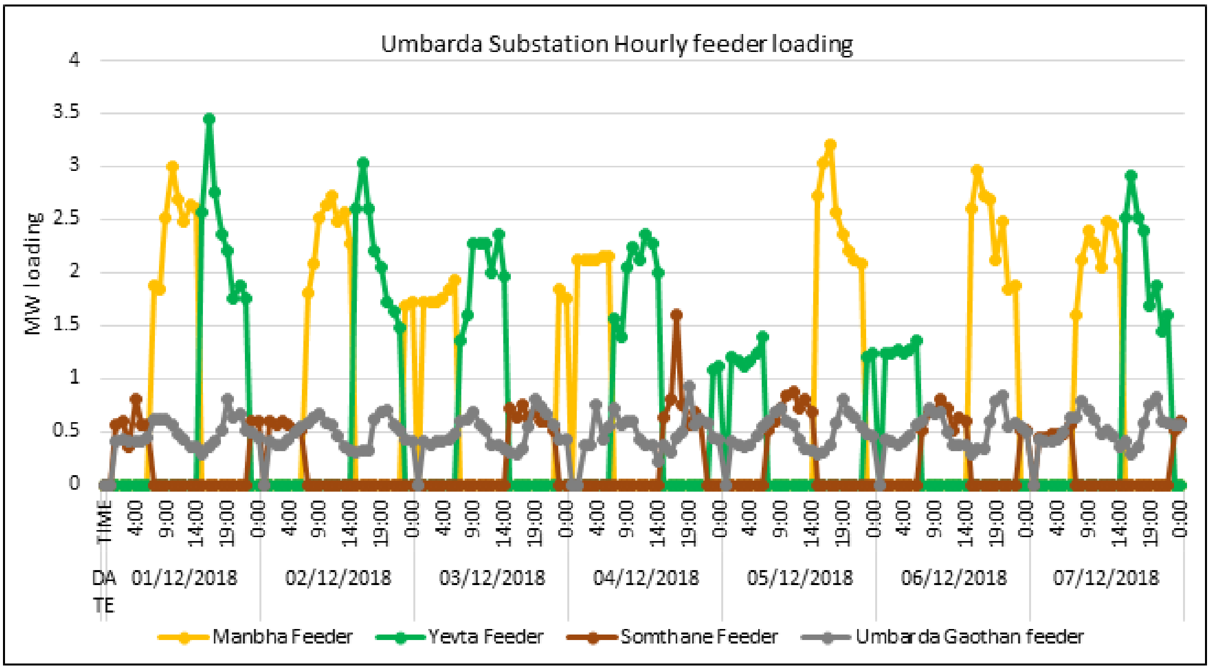


Figure 6: Feeder Loading during Peak period

Hence they are operated in 3 time slots: 7am-3pm, 3pm-11pm, 11pm -7am, each feeder getting the day supply 2 out of 6 days. Because this is not according to protocol, an additional 3 MVA transformer has been purchased. While the peak hours are avoided to reduce costs, there is a trade-off in infrastructure cost. A 3 MVA transformer costs ~ 70 lakhs, and instead the 3 time slots, with greater number of daytime supply, may result in Rs. 3.5 lakhs additional energy cost per year.

3.2 Establishing and improving design protocols

The LT network restructuring exercise needs further refinement using data on latent demand (unsanctioned connections on bore wells and extra pump sets on one connection), and a more detailed timeline of demand development. Other aspects like cost of restructuring, a simple and low-cost protocol to find the load diversity factor, are yet to be investigated.

3.3 Communities and Demand-Side Management

Demand-side management by communities is an important part of a good solution. A simple yet highly effective and low cost improvement would be capacitor usage by farmers. In our studies we find that farmers are not just unaware of the benefits of capacitor usage but in fact belabour under many misunderstandings. Awareness creation and drives must be conducted at a DT/village level in a way that the benefits are clearly visible to farmers. Other initiatives such as staggered usage on overloaded DTs, and staggering of start time of auto-starters, will require a stronger community involvement. However, with social media and Information Technology, community action can be made convenient and useful to all.

Given the controlled nature of the Ag feeders, a smart grid approach, combined with community intiatives, could lead to many benefits for MSEDCL as well as farmers.

Appendix I

Table 12: Work Estimate 19-20 Cost Data used for per farmer infrastructure cost calculation

Item Code No.	Work Description	DPR Cost (Lakhs) 19-20	Remarks
0101	33/11/ kV New S/S (Supply, erection, testing & commissioning) 1 x 5 MVA S/S	191.7	
0301	33/11 kV Additional Power Transformer (Supply, erection, testing & commissioning) 1*5MVA	95.7	3MVA addition data not available. 5MVA considered
0908	11kV, Pin type with 55 Sq. mm AAAC conductor on 9 mtr PSC poles	4.7	
1212	63 KVA Dist. Transformer centers on 9 mtrs 100 X 116 mm RSJ poles with Kit-kat Dist box.	2.8	
1213	100 KVA Dist. Transformer centers on 9 mtrs 100 X 116 mm RSJ poles with Kit-kat Dist box.	3.3	
1218	25 KVA DTC on RSJ, 110 X 116, 9 mtr with Kit Kat DB	1.9	
1401 **16-17 data*	3 Phase, 4 Wire LT line with ANT for Phase & GNAT for Neutral on RSJ 125 X 75, 8 mtr	2.7	Extrapolated from 16-17 data, not available in 19-20 data
Three core submersible cable		Rs 80/m	

Appendix II

Heuristics for restructuring

The paid pending consumers and the unconnected wells were first connected to the nearest LT line according to the method followed by MSEDCL. After adding the 14 Paid pending and 78 unconnected wells, 3 DTs exceeded their limit of a maximum number of pumps that can be added on a DT (24 for 100 kVA and 17 for 63 kVA). Hence these 92 consumers cannot be added according to the method followed by the utility to give the supply. To add all these consumers with minimal DT and LT line additions, we restructure the LT network as follows.

The two DTs that are overloaded are Ambedkar DT and Dighde DT. Three DTs which are close to the dam (Dharan1, Dharan2 and Somnathe DT) do not have an LT network as the connections on these DTs have service wires attached to a single pole. These 3 DTs are not considered for this analysis. The remaining 12 DTs can be distributed in 3 clusters for redistribution of loads, as shown in Figure 7.

- Ambedkar DT and its surrounding 4 DTs that are not overloaded.
- Dighde DT and its surrounding 5 DTs that are not overloaded.
- Sattar Babu DT.

Sattar Babu DT is far away from both of the clusters as mentioned above. The restructuring is done separately for the three clusters. This makes it easier when done manually. However, this method can be automated, and in that case clusters are not needed.

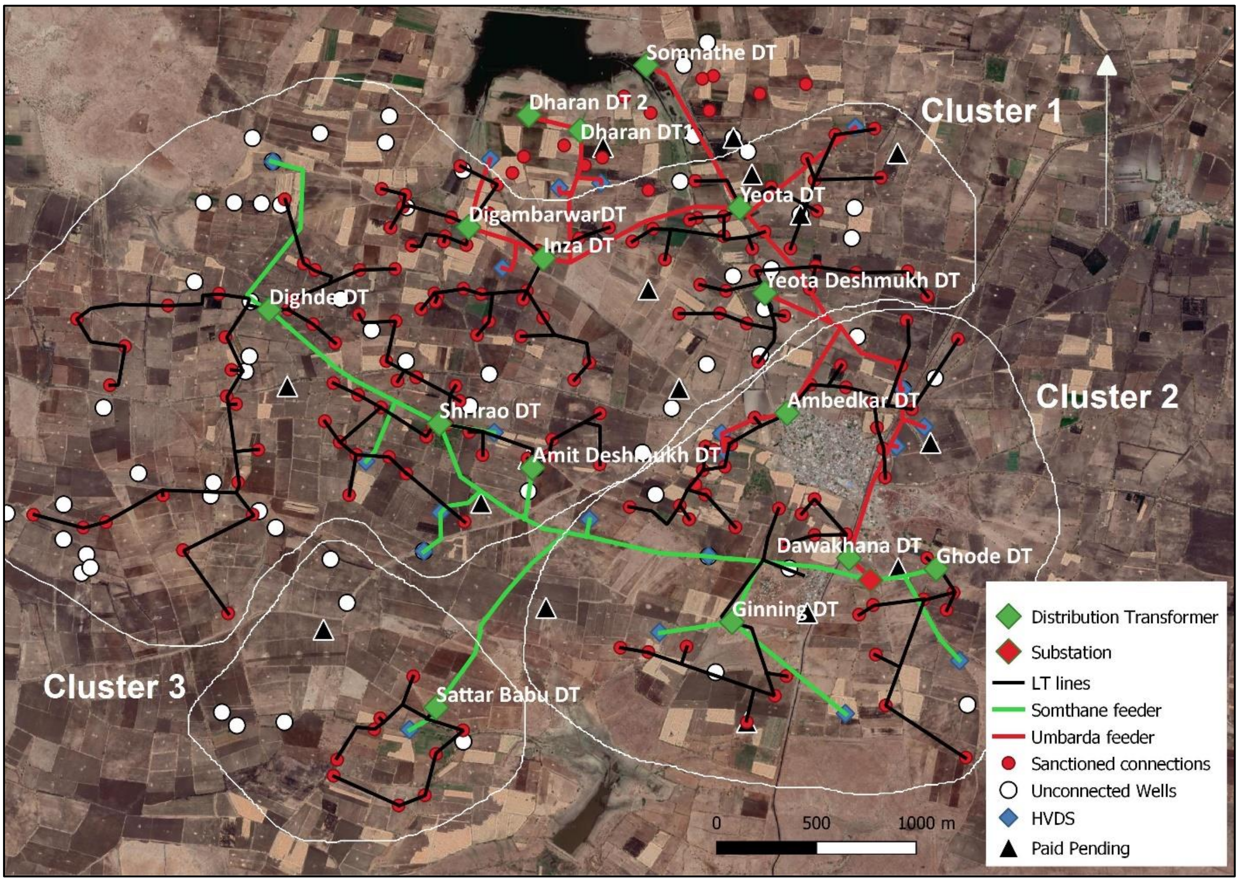


Figure 7: Three clusters identified for redistribution of LT network

The protocol for connecting loads within each cluster is as follows:

- Considering each DT as the starting point. Consider a circle of radius 300 m expanding in around each DT. The nearest pump(s) that come within the circle of a DT is (are) connected to the particular DT.
- The circles continues to expand. When a pump comes within a circle, it is connected to the closest point in the LT network of that DT. That point in the LT network may be the DT itself or a pole on the LT network of particular DT.
- If a pump that comes within a circle is already connected to an LT network it is ignored, and the circle continues to expand.
- While connecting any unconnected pump, the following constraints are considered:
  - If a pump is included in two circles at the same time, it is connected to the one which makes a shorter connection, or in other words, it is connected to the nearest LT line.
  - The maximum number of pumps that can be added on a DT should be within the prescribed limit of DT loading based on load diversity factor and DT rating.

    - If connecting the next pump means exceeding the limit as mentioned above, that pump is ignored by that circle, and it could get included in another circle.
    - The length of the LT lines cannot be extended beyond the limit that does not allow a voltage drop beyond the standard limit of 240-6%. This will depend on the length as well as the connected load on the LT line.
- At the end of this exercise if there are some unconnected loads, then some adjustments can be attempted by transferring loads across DTs so that these unconnected loads can be incorporated on the nearest LT lines.
- Additionally, if a pump is outside the 300 m radius circle from a pole on the LT network, but within the 300 m radius from an existing Paid Pending connection, then that pump has been taken into account.
- If even after this some loads are unconnected, a new DT may need to be installed.

These set of heuristics may not lead to a fully optimized solution, but they can be used to reach a practical and yet substantially improved solution. While the clusters make it easier to envision a solution, strictly speaking, it is not necessary to make clusters. An automated solution can be developed to make it easier to envision various possibilities and make it an easily implementable exercise on the basis of GIS mappings.

Appendix III

Table 5: Sanctioned loads, Paid pending pumps and unconnected wells

DT name*	**No of San. Con.**	**Paid Pending consumers**	**Unconnected wells seen from satellite image in the vicinity of the DT**	**Unconnected wells and Paid pending** *consumers that can be added by LT line extension*	**Probable illegal connections**
Ambedkar	25	1	7	0	2
Amit Deshmukh	16	2	9	11	5
Dawakhana	5	1	3	3	0
Digambarvar	13	0	7	7	2
Dighde	27	0	22	0	9
Ghode	9	1	2	3	0
Ginning (63kVA)	7	2	8	11	2
Inzha DT	18	0	2	2	0
Sattar Babu	9	1	5	6	1
Shirao	12	1	4	5	0
Yevta Deshmukh	13	2	8	10	4
Yevta Road	19	3	5	8	2
Somnathe	6	0	8	9	0
Dharan DT 1	5	3	1	4	0
Dharan DT2	5	0	1	1	0
Total	**189**	**17**	**92**	**80**	**27**

** All are 100 kVA DTs except Ginning DT*